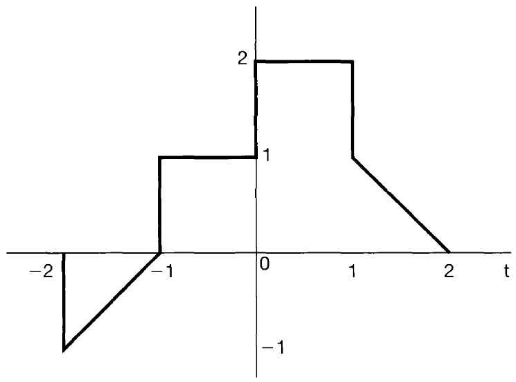
Homework 01

1.1. Express each of the following complex numbers in Cartesian form ($x + jy$): $\frac{1}{2}e^{j\pi}$, $\frac{1}{2}e^{-j\pi}$, $e^{j\pi/2}$, $e^{-j\pi/2}$, $e^{j5\pi/2}$, $\sqrt{2}e^{j\pi/4}$, $\sqrt{2}e^{j9\pi/4}$, $\sqrt{2}e^{-j9\pi/4}$, $\sqrt{2}e^{-j\pi/4}$.

1.2. Express each of the following complex numbers in polar form ($re^{j\theta}$, with $-\pi < \theta \leq \pi$): 5, -2, $-3j$, $\frac{1}{2} - j\frac{\sqrt{3}}{2}$, $1+j$, $(1-j)^2$, $j(1-j)$, $(1+j)/(1-j)$, $(\sqrt{2} + j\sqrt{2})/(1 + j\sqrt{3})$.

1.21 A continuous-time signal $x(t)$ is shown in Figure P1.21. Sketch and label carefully each of the following signals:

(a) $x(t-1)$ (b) $x(2-t)$ (e) $[x(t)+x(-t)]\,u(t)$



1.22 A discrete-time signal $x[n]$ is shown in Figure P1.22. Sketch and label carefully each of the following signals:

(a) $x[n-4]$ (b) $x[3-n]$ (e) $x[n]\,u[3-n]$